

THE MULTI-BAYESIAN SEQUENTIAL DECISION PROCEDURE

J.M. van Zyl. Department of Mathematical Statistics, University of the Orange Free State, P.O. Box 339,
and
D.J. de Waal BLOEMFONTEIN, Republic of South Africa.

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Summary: The methods of Weerahandi and Zidek for finding multi-Bayesian decision rules, based on the Kalai-Nash product, can be applied to a situation where there are, at most, M observations taken sequentially. Each Bayesian's utility for stopping and taking a decision, or continuing to take observations, after n have been taken, can be calculated recursively from the Bayesian's utilities for group decisions that would be made after $n+1$ observations. These utilities can then be used with the Kalai-Nash product to find group decisions after n observations. Hence, working backwards, the entire group decision rule can be found. An example is given.

1. INTRODUCTION

A group of Bayesians with different beliefs observe a sample sequentially of not more than M observations, from a population with density $f(x|\theta)$. A decision that depends on θ , the unknown state of nature, must be made co-operatively. The Kalai-Nash solution, as proposed by Weerahandi

and Zidek (1981)(1983) will be applied to find decision rules. Weerahandi and Zidek (1983) have discussed various solution concepts for multi-Bayesian statistical decision problems and presented criteria for the assessment of these methods. Using these criteria it is shown that the Kalai-Nash solution possesses advantageous properties for applications in multi-Bayesian decision problems.

Some of the favourable properties are: the solution remains invariant with respect to independent positive linear transformations of the utilities involved. This means that the utilities need not be comparable provided they are von Neumann-Morgenstern utilities; the rule is not invariant under more general monotonic transformations which still yield strategically equivalent value functions. If the Kalai-Nash rule is used, no Bayesian can be made to agree to a choice where he will suffer a net loss.

The concept of subsampling assessment has been defined by Weerahandi and Zidek (1983). This means that for a given group of Bayesians, only an unknown subgroup will implement the rule. The Kalai-Nash rule has the property that it will remain unchanged under these conditions.

The elements of the multi-Bayesian sequential procedure can be defined by analogy with the one-person problem as defined in Ferguson (1967) and Berger (1980) and are the following:

- (a) A group B of k Bayesians, $B = \{\beta_1, \dots, \beta_k\}$.

Let $\alpha(\beta)$ denote the weight of importance of β in the decision-making as specified for the Kalai-Nash solution $\sum_{\beta} \alpha(\beta) = 1$ and $\alpha(\beta) \geq 0$.

- (b) A - the space of actions and $A^* = AU\{a_0, a_c\}$.

The action space is enlarged to A^* to include a_c , where a_c denotes the action, take another observation (de Waal, et al. (1983)) in the sequential procedure.

Action a_0 means that no action will be taken and this will happen if no agreement can be reached. Each Bayesian will then retain his current utility at that stage.

(c) Θ - the parameter space and $\theta \in \Theta$ a parameter.

(d) $\underline{x}^n = (x_1, \dots, x_n)$ will denote a sample of size n with sample space $\mathcal{X}^n = \mathcal{X}_1 \times \dots \times \mathcal{X}_n$. We will consider examples where the x_i are identically and independently distributed. $\underline{x}^n$ has density function $f_n(\underline{x}^n | \theta) = \prod_{i=1}^n f_n(x_i | \theta)$ and distribution function $F_n(\underline{x}^n | \theta)$ on $\mathcal{X}^n$. β 's posterior density of θ after $\underline{x}^n$ has been observed, will be denoted by $\pi^n(\theta | \underline{x}^n, \beta)$, the distribution function by $\pi^n(d\theta | \underline{x}^n, \beta)$ and the prior density by $\pi(\theta | \beta)$.

The predictive density of $x_{n+1} | \beta$ after $\underline{x}^n$ has been observed, will be denoted by $m(x_{n+1} | \underline{x}^n, \beta)$ and the distribution function by $m(dx_{n+1} | \underline{x}^n, \beta)$ where

$$m(x_{n+1} | \underline{x}^n, \beta) = \int_{\Theta} f(x_{n+1} | \theta) \pi^n(d\theta | \underline{x}^n, \beta).$$

(e) The decision utility of β if action $a \in A$ is chosen and θ is the true state of nature will be denoted by $u(a | \theta, \beta)$, and $u_n(a | \theta, \underline{x}^n, \beta)$ is the total utility of β if action a is taken after $\underline{x}^n$ has been observed for a given θ . The total utility will be the utility of β resulting if action a is chosen less the cost of observing $x_1, \dots, x_n$. It will be assumed that $u(a | \theta, \beta)$ is bounded from above and $u(a | \theta, \beta)$ is defined for all $(\theta, a, \beta) \in \Theta \times A \times B$.

The reference utility level (Weerahandi and Zidek (1983)) of β is denoted by $C_N(\underline{x}^n, \beta)$. If no agreement can be reached, action a_0

will be taken and each $\beta \in B$ will retain his current utility level $C_N(\tilde{x}^n, \beta)$.

- (f) A cost function $c(\tilde{x}^n, \beta)$ where $c(\tilde{x}^n, \beta)$ is β 's cost of observing $x_1, \dots, x_n$. The cost of taking an observation is often a constant $c(\beta)$, independent of the observed values. In such a case

$$u_n(a | \theta, \tilde{x}^n, \beta) = u(a | \theta, \beta) - nc(\beta).$$

The Bayesians will choose an action at random, according to the randomized decision rule $\tilde{\delta}$, $\tilde{\delta} \in D$, where D is the space of randomized decisions, and this rule is found by using the Kalai-Nash method. The expected stop utility of β if an action $a \in A$ is chosen at stage n is given by

$$(1.1) \quad B(a | \tilde{x}^n, \beta) = \int_{\Theta} u_n(a | \theta, \tilde{x}^n, \beta) \pi^n(d\theta | \tilde{x}^n, \beta).$$

Let $B(a_c | \tilde{x}^n, \beta)$ denote the expected utility of β after observing $\tilde{x}^n$. This continuation utility can be defined recursively in terms of β 's expected utility after the next observation.

The Bayesians will randomize at a certain stage over the action space $A^* = AU\{a_0, a_c\}$. This means that at each stage a final action or another observation can be taken, conditional on $\tilde{x}^n$, the observations and the stopping time. The Bayesians will decide according to the decision rule $\tilde{\delta}(\cdot | \tilde{x}^n)$, where $\tilde{\delta}(da | \tilde{x}^n)$ is a probability distribution on A^* for given $\tilde{x}^n$. The expected utility of β if the decision rule is $\tilde{\delta}(\cdot | \tilde{x}^n)$, is

$$(1.2) \quad B(\tilde{\delta} | \tilde{x}^n, \beta) = \int_{a \in A^*} B(a | \tilde{x}^n, \beta) \tilde{\delta}(da | \tilde{x}^n).$$

The optimal decision rule $\tilde{\delta}_0(\cdot | \tilde{x}^n)$ is found by optimizing the Kalai-Nash product

$$(1.3) \quad N(\delta_0(\cdot | \tilde{x}^n)) = \prod_{\beta \in B} [B(\delta | \tilde{x}^n, \beta) - C_N(\tilde{x}^n, \beta)]_+^{\alpha(\beta)}$$

$$[B(\delta | \tilde{x}^n, \beta) - C_N(\tilde{x}^n, \beta)]_+ = \max(B(\delta | \tilde{x}^n, \beta) - C_N(\tilde{x}^n, \beta), 0).$$

This condition ensures that each Bayesian will receive no less than his current utility at that stage.

The continuation utility calculated recursively in terms of β 's expected utility after the next observation is

$$(1.4) \quad B(a_c | \tilde{x}^n, \beta) = \int_{x_{n+1}} B(\delta_0 | \tilde{x}^{n+1}, \beta) m(dx_{n+1} | \tilde{x}^n, \beta).$$

The action space is $AU\{a_0\}$ at stage M in the truncated procedure. The expected utilities can be calculated for given $\tilde{x}^n$ and the decision rules can be found. The continuation utilities can be calculated at stage $M-1$ by using formula (1.4) for given $\tilde{x}^{M-1}$. The action space is A^* and the decision rules can be found for all $\tilde{x}^{M-1}$. The optimal sequential procedure is constructed by continuing in an optimal way to stage 0. Thus a complete decision rule, δ , defining all future actions can be specified:

$$\delta \triangleq \{\tilde{\delta}(\cdot | \tilde{x}^1); \tilde{\delta}(\cdot | \tilde{x}^2); \dots; \tilde{\delta}(\cdot | \tilde{x}^M)\}.$$

2. EXAMPLE: MULTIPLE DECISION PROBLEMS INVOLVING THE PARAMETERS OF A MULTINOMIAL DENSITY

A decision must be taken that depends on the unknown parameter θ of a multinomial density. Suppose there are s categories and a sequential sample of not more than M can be taken at a certain cost for each Bayesian, from a population with density $f(\tilde{x} | \theta)$. Let $\Theta = \{\theta_1, \dots, \theta_p\}$

$\theta_i = (\theta_{i1}, \dots, \theta_{is})$, $\sum_j \theta_{ij} = 1$. Let $X_i = (X_1, \dots, X_s)$. If the outcome belongs to category j , $X_j = 1$, otherwise $X_j = 0$.

Suppose n observations have been made. Let r_j denote the number of times that outcome j occurred. The density of $\underline{r}_n = (r_1, \dots, r_s)$,

$\sum_{i=1}^s r_i = n$, is

$$(2.1) \quad f(\underline{r}_n | \underline{\theta}) = \frac{n!}{r_1! \dots r_s!} \theta_1^{r_1} \dots \theta_s^{r_s}.$$

Let $A = \{a_1, \dots, a_q\}$ and π_{ij} denote the prior probability that

$\underline{\theta} = \underline{\theta}_i$ for β_j . The decision utility of β_j if $\underline{\theta} = \underline{\theta}_i$ and action a_ℓ is chosen equals $u_{\ell ij} \geq 0$. The various decision utilities can be summarized in a matrix $U(\beta_j)$ where $[U(\beta_j)]_{\ell k} = u_{\ell kj}$.

The posterior density of $\underline{\theta}_i$ for β_j after $\underline{r}_n$ has been observed is

$$(2.2) \quad \pi^n(\underline{\theta}_i | \underline{r}_n, \beta_j) = \frac{\pi_{ij} \theta_{i1}^{r_1} \dots \theta_{is}^{r_s}}{\sum_{i=1}^p \pi_{ij} \theta_{ij}^{r_1} \dots \theta_{is}^{r_s}}.$$

The predictive density of X_{n+1} is

$$(2.3) \quad m(x_{n+1} | \underline{r}_n, \beta_j) = \frac{\sum_{i=1}^p \pi_{ij} \theta_{i1}^{r_1+x_1} \dots \theta_{is}^{r_s+x_s}}{\sum_{i=1}^p \sum_{x_{n+1} \in X_{n+1}} \pi_{ij} \theta_{ij}^{r_1+x_1} \dots \theta_{is}^{r_s+x_s}}.$$

If action a_ℓ is taken at stage n , the expected utility of β_j is

$$(2.4) \quad B(a_\ell | \underline{r}_n, \beta_j) = \sum_{i=1}^p \pi^n(\underline{\theta}_i | \underline{r}_n, \beta_j) u_{\ell ij} - c(\underline{r}_n, \beta_j).$$

Let $B(a_c | \underline{r}_n, \beta)$ denote β 's expected utility if the Bayesians decide to take another observation at stage n . The decision problem at stage n , after $\underline{r}_n$ has been observed, is summarized in table 3.1.

	a_1	a_2	$\dots$	a_q	a_c
β_1	$B(a_1 \underline{r}_n, \beta_1)$	$B(a_2 \underline{r}_n, \beta_1)$	$\dots$	$B(a_q \underline{r}_n, \beta_1)$	$B(a_c \underline{r}_n, \beta_1)$
β_2	$B(a_1 \underline{r}_n, \beta_2)$	$B(a_2 \underline{r}_n, \beta_2)$	$\dots$	$B(a_q \underline{r}_n, \beta_2)$	$B(a_c \underline{r}_n, \beta_2)$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
β_k	$B(a_1 \underline{r}_n, \beta_k)$	$B(a_2 \underline{r}_n, \beta_k)$	$\dots$	$B(a_q \underline{r}_n, \beta_k)$	$B(a_c \underline{r}_n, \beta_k)$

Table 3.1: Expected utilities of $\beta_1, \dots, \beta_k$, after $\underline{r}_n$ has been observed, for the various actions.

Let $C_N(\underline{r}_n, \beta)$ denote β 's reference utility. The optimal decision rule δ_0 is the value $\delta_0 = (\delta_1, \dots, \delta_q, \delta_{q+1})$ that maximizes:

$$(2.5) \quad N(\delta) = \prod_{\beta \in B} \left[\sum_{i=1}^q \delta_i B(a_i | \underline{r}_n, \beta) + \delta_{q+1} B(a_c | \underline{r}_n, \beta) - C_N(\underline{r}_n, \beta) \right]_+^{\alpha(\beta)} \quad \sum_{i=1}^{q+1} \delta_i = 1, \delta_i \geq 0.$$

Action a_i will be taken with probability δ_i . The expected utility of β at stage n if this procedure is used and agreement has been reached is

$$(2.6) \quad B(\delta_0 | \underline{r}_n, \beta) = \sum_{i=1}^q \delta_i B(a_i | \underline{r}_n, \beta) + \delta_{q+1} B(a_c | \underline{r}_n, \beta)$$

(or else $C_N(\underline{r}_n, \beta)$ if no solution exists such that every Bayesian will gain over his current utility at that stage).

$B(a_c | r_{n-1}, \beta)$ calculated in terms of the expected utilities at state n is given by

$$(2.7) \quad B(a_c | r_{n-1}, \beta) = \sum_{x_n \in x_n} m(x_n | r_{n-1}, \beta) B(\delta_0 | r_{n-1} + x_n, \beta).$$

Example 3.1:

β_1 and β_2 must decide between actions a_1 and a_2 and their choice is influenced by the value of the parameter of a Bernoulli density. The parameter θ is one of three values $\theta_1 = 0,9$, $\theta_2 = 0,1$ or $\theta_3 = 0,5$. The decision utilities are

		θ_1	θ_2	θ_3
β_1	a_1	5	1	2
	a_2	3	3	2

Table 3.1: Decision utilities of β_1 if an action a is taken and θ is the parameter value.

		θ_1	θ_2	θ_3
β_2	a_1	3	2	2
	a_2	2	4	3

Table 3.2: Decision utilities of β_2 if an action a is taken and θ is the parameter value.

The prior probabilities of β_1 are $\pi_{11} = \frac{1}{3}$, $\pi_{21} = \frac{1}{3}$ and $\pi_{31} = \frac{1}{3}$, and β_2 's probabilities are: $\pi_{12} = \frac{1}{4}$, $\pi_{22} = \frac{1}{4}$ and $\pi_{32} = \frac{1}{2}$. Not

more than $M=2$ observations can be taken from a Bernoulli population with parameter θ .

β_1 's utility will decrease by 0,04 per observation and it will cost β_2 0,1 to take an observation. Thus $c(r_n, \beta_1) = 0,04$ and $c(r_n, \beta_2) = 0,1$. Suppose that $\alpha(\beta_1) = 0,6$, $\alpha(\beta_2) = 0,4$ and $C_N(r_n, \beta) = (M-n)c(r_n, \beta)$ for β_1 and β_2 . $A = \{a_1, a_2\}$ and $\Theta = \{\theta_1, \theta_2, \theta_3\}$.

A decision triangle similar to that of Berry and Wang (1982) can be constructed. The decision of β_1 is displayed in the upper left corner, that of β_2 in the upper right corner and their co-operative decision in the lower part of each block.

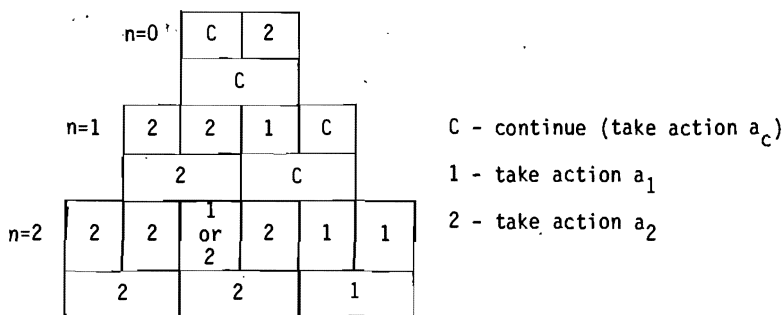


Figure 3.1: Decision triangle for Bayesians β_1 and β_2 . Horizontal blocks represent $r_n = 0, 1, \dots, n$.

Note that no randomized rules had to be used.

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